Week 36, Tuesday Session

Flipped classroom: your questions, then bias-variance and cross-validation in practice

Bring your laptop – we work in `week36tuesday.ipynb`

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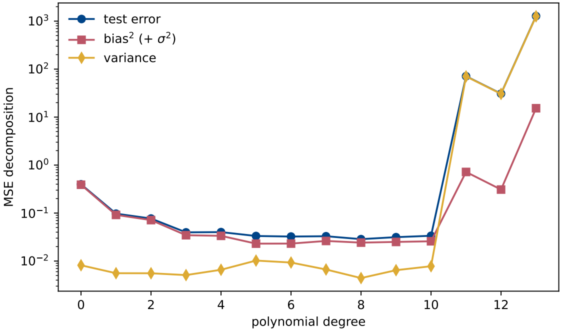
How today works

The format (as every Tuesday)

Mondays are lectures; Tuesdays belong to your questions and to concrete cases. You have heard the lecture and skimmed Sections 2.3–2.12.

1. **First $\sim$20 min:** open questions from Monday, then the warm-ups on the next slides, discussed with your neighbours and in plenum.
2. **Rest of the double session:** *two* cases only this week – the bias-variance decomposition, and cross-validation – worked in pairs or small groups from `week36tuesday.ipynb`. Deep beats broad: the goal is that you leave owning Eq. (2.47) and a working k-fold loop, both of which Project 1 asks for.
3. **Last 5 min:** wrap-up, exercise deadline, and what next Monday builds on.

Picking up from Monday: the measured decomposition



Monday's lecture closed the loop on last week's U-curve: the test error is, by Eq. (2.47), the sum

$$\mathrm{Bias}^2 + \mathrm{Var} + \sigma^2,$$

and the bootstrap let us *measure* each term. Bias falls and flattens, variance explodes, the sum has a broad minimum (degrees 5–9 here). Today you reproduce this figure

yourselves, stress-test it – more data, more noise, a Ridge penalty – and then let cross-validation find the minimum without ever touching the test set.

Warm-up 1: the three terms, in your own words

Question

Without looking at your notes: which term of Eq. (2.47) does each sentence describe? (a) “No amount of data fixes a model that is too simple.” (b) “Redraw the training set and the prediction jumps.” (c) “Even the perfect model gets this wrong.”

Warm-up 1: the three terms, in your own words

Question

Without looking at your notes: which term of Eq. (2.47) does each sentence describe? (a) “No amount of data fixes a model that is too simple.” (b) “Redraw the training set and the prediction jumps.” (c) “Even the perfect model gets this wrong.”

Answer

(a) the squared bias – the error of the *average* prediction, built into the model's assumptions; (b) the variance – sensitivity to the particular training sample; (c) the irreducible σ^2 – the noise no model predicts. Bonus check: which one does Ridge attack? The variance, at the price of some bias.

Warm-up 2: read the code, not the formula

Question

In Monday's bootstrap code the bias is computed as `np.mean((y_test - np.mean(y_pred, axis=1, keepdims=True))**2)`. Two questions: why `axis=1`, and why does this "bias" contain σ^2?

Warm-up 2: read the code, not the formula

Question

In Monday's bootstrap code the bias is computed as `np.mean((y_test - np.mean(y_pred, axis=1, keepdims=True))**2)`. Two questions: why `axis=1`, and why does this "bias" contain σ^2?

Answer

Columns of `y_pred` are bootstrap training sets, so averaging over `axis=1` is the expectation over training sets, $\mathbb{E}[\tilde{y}]$ – averaging over `axis=0` would average over test points instead, which is meaningless here. And the code compares with $y_{\text{test}} = f + \varepsilon$, not the unknown f: the noise cannot be subtracted out, so the measured bias absorbs σ^2 and the printed identity is really an inequality.

Warm-up 3: predict before you compute

Question

Case 1 below reruns the bias-variance sweep with (a) $n = 400$ instead of 40 data points, and (b) noise $\sigma = 0.3$ instead of 0.1. Predict: what happens to each of the three curves in each case?

Warm-up 3: predict before you compute

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Answer

(a) More data tames the variance (roughly like $1/n$ for fixed complexity): the explosion moves to higher degree and the minimum shifts right – with more data you can afford a richer model. The bias curve is unchanged: it is a property of the model family, not of n. (b) More noise lifts the floor (σ^2 is 9 times larger) and scales up the variance term; the bias^2 term is unchanged, but the minimum moves *left*: noisier data favour simpler models. Now go check.

Case 1: the bias-variance decomposition, owned

Goal: reproduce Monday's figure, then break it and understand the pieces (notebook, Case 1; seed 2026 gives exactly the lecture numbers).

1. Run the sweep ($n = 40$, 100 bootstraps, degrees 0–13) and reproduce `week36_bias_variance.pdf`.
2. **Vary the data size:** $n = 100$ and $n = 400$. Where does the variance explosion move? Does the bias curve care?
3. **Vary the noise:** $\sigma = 0.3$. Confirm your warm-up 3 prediction for floor and minimum.
4. **Swap the model:** replace OLS by Ridge with $\lambda \in \{10^{-4}, 10^{-1}, 10\}$ at fixed high degree 12. Watch the variance fall and the bias rise as λ grows – the tradeoff turned into a dial.

Checkpoints (plenary after ∼25 min)

Degree 0 has bias$^2 \approx 0.39$ ($\approx$ the variance of f itself – why?); the minimum is broad, degrees 5–9; at degree 13 the variance is $\sim 10^3$. For step 4: which λ gives the smallest *total* error at degree 12, and is it better than plain OLS at degree 6?

Case 2: cross-validation: choosing λ without touching the test set

Goal: a k-fold loop you wrote yourself, validated against `scikit-learn`, then used for real model selection (notebook, Case 2).

1. Implement 5-fold CV for Ridge on $y = 3x^2 + \varepsilon$ (degree-6 polynomial, λ on `np.logspace(-3, 5, 100)`), scaler fitted *inside* each fold. Reproduce `week36_cv_ridge.pdf`.
2. Check yourself against `cross_val_score` with a `Pipeline`: the two curves must coincide to plotting accuracy. If they do not, the usual culprit is scaling fitted on the full data.
3. Read off λ^* and compare the minimal CV-MSE with the known $\sigma^2 = 1$. Vary $k \in \{5, 10, 20\}$: how stable are λ^* and the error bar (the spread over folds)?
4. **Close the week's loop:** run CV over the polynomial *degree* for the week-35 data set (OLS, degrees 0–13) and compare the CV curve with Case 1's test-error curve – same minimum, no test set consumed.

Checkpoints

With seed 2026: $\lambda^* \approx 1.2$, CV-MSE ≈ 1.5 (just above the noise floor). For step 4: CV picks a degree in the same broad 5–9 window as the bootstrap decomposition – two independent methods, one answer.

Wrap-up, deadlines, and Monday

- **What you should own now:** Eq. (2.47) and what each term means operationally; a bootstrap loop; a leak-free k-fold loop with the preprocessing inside the folds.
- **Weekly exercises** (`exercisesweek36.ipynb`): the analytical steps of the bias-variance derivation, plus bootstrap and cross-validation on your own code. Deadline **Friday September 4 at midnight**.
- **Project 1** asks for exactly today's machinery (bias-variance analysis and cross-validation on regression models) – the code you wrote today is directly reusable.
- **Next Monday and week 38:** the main focus shifts to *optimization*: gradient descent methods – from plain and steepest descent via momentum to stochastic gradient descent – the workhorse behind everything from Lasso to deep networks. We will also briefly mention the Bayesian reading of OLS, Ridge and Lasso (a Gaussian prior gives Ridge, a Laplace prior the Lasso).

Before you leave

One line on a piece of paper (or the digital board): the muddiest point of the week – what should Monday's first ten minutes revisit?